## C3.7 Snoop response channel signaling

Table C3-20 shows the signals on the snoop response channel.

**Table C3-20 Snoop response channel signals**

| Signal | Source | Description |
|---|---|---|
| **CRVALID** | Master | Snoop response valid. This signal indicates that the channel is signaling a valid snoop response. |
| **CRREADY** | Interconnect | This signal indicates that the snoop response can be accepted in the current cycle. |
| **CRRESP[4:0]** | Master | Snoop response. This signal indicates the response to a snoop transaction and how the transaction completes. |

The standard AXI **VALID**/**READY** handshake rules apply. See *Handshake process* on page A3-37.

When the **CRVALID** signal is asserted the snoop response on **CRRESP**, must not change until the interconnect sets **CRREADY** HIGH. When **CRVALID** is asserted, it must remain asserted until **CRREADY** is asserted.

It is permitted to assert **CRREADY** before or in the same cycle as **CRVALID**. If **CRREADY** is asserted before **CRVALID** then **CRREADY** can be deasserted without **CRVALID** being asserted.

A snoop response is required on the snoop response channel for each snoop address that is presented to a cached master on the snoop address channel.

All snoop transactions are ordered. A response on the snoop response channel must be in the same order as the addresses on the snoop address channel

Table C3-21 shows the allocation of each bit of **CRRESP[4:0]**.

**Table C3-21 Snoop response bit allocations**

| CRRESP[4:0] | Meaning |
|---|---|
| [0] | DataTransfer |
| [1] | Error |
| [2] | PassDirty |
| [3] | IsShared |
| [4] | WasUnique |

The meaning of the snoop response bits and the limitations of use are as follows:

**CRRESP[0], DataTransfer bit**

When HIGH, it indicates that a full cache line of data will be provided on the snoop data channel for this transaction.

When LOW, it indicates that no data will be provided on the snoop data channel for this transaction.

The snoop transaction type and the state of the cache line in the snooped cache determine if the DataTransfer bit is set HIGH and a data transfer occurs.


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For the following transactions, a data transfer occurs if the snoop process has resulted in a cache hit:

- ReadOnce
- ReadClean
- ReadNotSharedDirty
- ReadShared
- ReadUnique.

If the cache line is clean, it is not mandatory that data transfer occurs. However, this specification recommends that data transfer still occurs.

For the following transactions, data transfer must occur if the snoop process has resulted in a cache hit and the cache line is dirty:

- CleanInvalid
- CleanShared.

A MakeInvalid transaction never requires a data transfer.

The DataTransfer bit can be set to 1 and data can be returned on the snoop data channel when it is not required. For example, a CleanInvalid transaction can return data when it holds the cache line in a Clean state, and a MakeInvalid transaction can return data. However, this specification does not recommend this use of the snoop data channel.

**Note**

The protocol permits the return of data on the snoop data channel when it is not required to enable a simplified snoop port implementation to handle a MakeInvalid transaction in the same manner as a ReadUnique or CleanInvalid, and a CleanInvalid transaction to be handled in the same manner as a ReadUnique.

**CRRESP[1], Error bit**

When HIGH, it indicates that the snooped cache line is in error. Typically, this is caused by a corrupt cache line that has been detected through the use of an *Error Correction Code*, (ECC) system.

If an error is detected, the snooped cache can take appropriate action, such as discarding a clean cache line. Alternatively, the snooped cache can flag the error by setting the Error bit HIGH and take no further action.

**Note**

The use of an error response is optional in the ACE protocol. Components that do not support error responses must permanently set the Error bit LOW.

**CRRESP[2], PassDirty bit**

When HIGH, it indicates that before the snoop process, the cache line was held in a Dirty state and the responsibility for writing the cache line back to main memory is being passed to the initiating master or the interconnect.

For all transactions, except MakeInvalid, if the cache line was held in a Dirty state before the snoop process and a copy is not being retained by the cache, then the PassDirty bit must be set HIGH.

For the following transactions, the responsibility for writing the dirty cache line back to main memory can be passed to the master requesting the data:

- ReadNotSharedDirty
- ReadShared
- ReadUnique.

In other cases, such as ReadClean, the dirty cache line must be written back to main memory by the interconnect.


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**CRRESP[3], IsShared bit**

When HIGH, it indicates that the snooped cache retains a copy of the cache line after the snoop process has completed.

The restrictions on the use of IsShared are:

- For the following transactions, the cache line in the snooped cache must be invalidated and the IsShared response must be LOW:
  - ReadUnique
  - CleanInvalid
  - MakeInvalid.
- For the following transactions, the snooped cache can determine if it retains a copy of the cache line after the snoop process has completed and the snooped cache must use the IsShared bit to signal the outcome:
  - ReadOnce
  - ReadClean
  - ReadNotSharedDirty
  - ReadShared
  - CleanShared.

Typically, a snooped cache retains a local copy of the cache line after the snoop process has completed. However, there are cases when a snooped cache does not retain a local copy, such as when passing the cache line as unique to another cache in response to a ReadNotSharedDirty snoop transaction.

**Note**

For a ReadOnce snoop transaction, the IsShared bit must be set to 1 if the snooped master is retaining a copy of the cache line, even if it is keeping the line in a Unique state.

**CRRESP[4], WasUnique bit** 表示这个CacheLine在被Snoop之前是Unique状态的

When HIGH, it indicates that the cache line was held in a Unique state before the snoop process.

The WasUnique bit must only be HIGH if it is known that no other cache can have a copy of the cache line.

A WasUnique response permits the snoop process to be terminated because no other cache can hold a copy of the cache line. WasUnique可以用于提前中止Snoop的进行，因为已经知道其他的设备不会拥有这个CacheLine

The protocol permits a cache to not generate a WasUnique response. In this case, the WasUnique bit must be permanently LOW. 协议上也允许WasUnique永远不会产生，那么这时候这个Bit就必须一直保持为0

**Note**

Permanently setting the WasUnique LOW can result in the cache line being provided to the original requester as Shared when it could have been provided as Unique. It can also result in additional caches being needlessly snooped.

Table C3-22 on page C3-179 shows the **CRRESP** response meanings and transactions for which they are valid.


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Table C3-22 does not show the Error bit **CRRESP[1]**, of the response field, because the value of this bit does not affect the meaning of the other snoop response bits. In this table:

- The meaning of the bits in the **CRRESP** response fields, when asserted, are:

  **WU, WasUnique** The cache line was in Unique state before this snoop.

  **IS, IsShared** The cache is keeping a copy of this cache line after this snoop.

  **PD, PassDirty** The cache line was dirty before this snoop. This response transfers responsibility for updating main memory, as well as the data.

  **DT, DataTransfer** The response to the snoop transaction includes a transfer on the snoop data channel.

- The snoop transactions are abbreviated as follows:

  **RO** ReadOnce
  **RC** ReadClean
  **RN** ReadNotSharedDirty
  **RS** ReadShared
  **RU** ReadUnique
  **CI** CleanInvalid
  **MI** MakeInvalid
  **CS** CleanShared.

- Whether a response is permitted, for each transaction, is indicated as follows:

  **E** Expected response
  **P** Permitted response
  **No** Response not permitted.

**Table C3-22 Response meanings and transactions for which they are valid**

| CRRESP[3:2,0] [4] | | | | Snoop transaction | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **IS** | **PD** | **DT** | **WU** | **RO** | **RC** | **RN** | **RS** | **RU** | **CI** | **MI** | **CS** | **Response meaning** |
| 0 | 0 | 0 | 0 | E | E | E | E | E | E | E | E | Line was invalid or has been invalidated. |
| 0 | 0 | 0 | 1 | P | P | P | P | P | E | E | E | Line was unique but has been invalidated. |
| 0 | 0 | 1 | x | P | P | P | P | E | P | P | P | Passing clean data before invalidating. |
| x | 1 | 0 | x | No | No | No | No | No | No | No | No | Cannot assert PassDirty with DataTransfer low. |
| 0 | 1 | 1 | x | P | E | E | P | E | E | P | P | Passing dirty data before invalidating. |
| 1 | 0 | 0 | x | P | P | P | P | No | No | No | E | Line is valid and clean but not being passed. |
| 1 | 0 | 1 | x | E | E | E | E | No | No | No | P | Passing clean data and keeping copy. |
| 1 | 1 | 1 | x | P | E | E | E | No | No | No | E | Passing dirty data and keeping copy. |

The following responses are illegal:

- IsShared, **CRRESP[3]** = 1 for:
  - ReadUnique
  - CleanInvalid
  - MakeInvalid.
- PassDirty, **CRRESP[2]** = 1, and DataTransfer, **CRRESP[0]** = 0, for any transaction.


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